POCKET AI ASSISTANT · FINAL BUILD · EVERY PART, WITH ITS LISTING

Everything the full build needs

Harness, battery, charging, frame and the tools to make them. Each row links to the listing you bought it from, or the one to buy. The whole device needs **one purchase**: the PowerBoost. One thing to check: a micro-USB cable for charging.

SUMMARY

38 items · 0 to buy · everything else is on the bench. The rechargeable battery is the 1200 mAh cell you own; it charges through the PowerBoost's USB port, in the device, with any 5 V USB charger.

The five modules

PART	EXACTLY WHICH		ROLE
ESP32-C3 SuperMini (A1)	Meshnology, 10-pack	`OWN`	the brain; 1 used
0.96" OLED, SSD1306	Hosyond, 5-pack	`OWN`	the eyes; reads SDA·SCL·VCC·GND pins-at-bottom
INMP441 microphone	AITRIP, 5-pack, round Ø13	`OWN`	lies chip-side down, hole up
MAX98357A amplifier	3-pack, terminal block opposite the pins	`OWN`	pins face the channel, VIN at the front
Speaker, 8 Ω 0.8 W, enclosed	Same Sky CES-20134-088PM	`OWN`	its own leads into the block, never cut
Tactile button	QTEATAK kit	`OWN`	on the USB socket's Kapton

Small parts

PART	EXACTLY WHICH		ROLE
Resistors: 4.7 kΩ (or 1 kΩ), 10 kΩ	LuminologyPro kit	`OWN`	R1, R3; measure each (R2 GAIN and R4 mic-SD dropped 2026-10-14)
220 µF electrolytic	ALLECIN kit	`OWN`	C1, stripe to GND
100 nF ceramic	BOJACK kit	`OWN`	C2, at the amp's VIN/GND stubs

Power and battery

PART	EXACTLY WHICH		ROLE
PowerBoost 1000C (third-party copy)	sold as 'For Powerboost 1000 … 2465 Module'; charger + load sharing + 5 V boost, EN switch pin — verify on the bench	`OWN`	the one board that owns the power; micro-USB charge port

1200 mAh LiPo, protected, JST-PH	Adafruit #258	`OWN`	the only owned cell rated for the load; 62 × 34 × 5
Slide switch	Chanzon SPDT mini, 25-pack	`OWN`	on the EN pin; carries no current
26 AWG silicone wire	TUOFENG	`OWN`	5 V and GND from the PowerBoost
30 AWG silicone wire	CBAZY	`OWN`	EN, LBO
Micro-USB cable + any 5 V USB charger	the PowerBoost's port is micro-USB	`CHECK`	if you want USB-C instead: Seeed LiPo Rider Plus in place of the PowerBoost, same wiring

Wiring and frame, brass build

PART	EXACTLY WHICH		ROLE
1.0 mm brass rod	K&S #9861, 5 per pack (10 owned)	`OWN`	rails and all 19 rods
1.5 mm brass tube	K&S #9831, 4 per pack	`OWN`	the frame loops; posts from the rod
Harris Stay-Clean acid flux	SCLF4	`OWN`	brass tube frame only, wash the same day
Jeweler's saw	SE 3-in-1	`OWN`	the tube; never cutters

Wiring and frame, silver build (the one-frame battery version)

PART	EXACTLY WHICH		ROLE
18 ga .925 sterling round wire, 5 ft	WireJewelry, dead-soft	`OWN`	rails and rods; straighten each piece first
14 ga .925 sterling round wire, 5 ft	WireJewelry, dead-soft	`OWN`	two loops at 65 × 39 inside, four posts of 35

Consumables

PART	EXACTLY WHICH		ROLE
Cardstock, 65 lb	Amazon Basics	`OWN`	jig card, shelf, cell cradle
Kapton tape	ELEGOO	`OWN`	socket roof, amp strip, A1's port, every tube
Heat-shrink, 1.5–2 mm	Pointool kit, white	`OWN`	5 V rod, GND rod, SCL, the 220 µF leads
63/37 solder, 0.8 mm	MAIYUM	`OWN`	
No-clean flux pen	Chip Quik CQ4LF	`OWN`	every electronics joint
Solder wick	JoTownCand	`OWN`	bridges between stubs
400 grit wet/dry	BAISDY	`OWN`	stubs and joint spots bright
91 % isopropyl alcohol	Epic Medical Supply	`OWN`	wipe after sanding paint
Hot glue, low temperature	SHJADE mini gun + sticks	`OWN`	six dots at the very end; no link recorded

Tools

PART	EXACTLY WHICH		ROLE
Soldering station + silicone mat + helping hands + tweezers	X-Tronic 3020-XTS kit	`OWN`	no link recorded; electronics tip for the harness
Soldering tips	900M-T assortment + tinner	`OWN`	no link recorded
Flush cutters	BOENFU 6"	`OWN`	pins, rod, leads; never the tube
Chain-nose + round-nose pliers	WORKPRO 3-pack	`OWN`	the strip off the headers, every bend
Digital caliper	NEIKO 01407A	`OWN`	stubs 1.5–2.5, the 12.5 mm screen block
Multimeter	KAIWEETS TRMS	`OWN`	continuity after every move
Heat gun	QWORK mini	`OWN`	no link recorded; shrinks the sleeves, never near the mic
Cutting mat	Dahle Vantage 10672	`OWN`	cutting the cards; never solder on it

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Sources. Ownership and listings from `docs/INVENTORY.md` (Amazon #1–#19, DigiKey, Adafruit), the two capacitor kits from `docs/MATERIALS.md`. "No link recorded" means the purchase is recorded without its listing. The PowerBoost is the only part not yet bought.